Parameters and architecture

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Teach the count with its assumptions.

Show every incoming/outgoing connection and receiving bias;separate parameter values,computed activations and training settings.

Keep one habit: Assess both layers of the independent architecture and the stated evidence limits using checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Distinguish architectural choices from **parameter** values

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Scores, activations and **output**s may change. The weights, biases, connections and activation choices stay fixed.

Opening retrieval

Without opening the old lesson, explain one key idea from Forward passes by hand. Give an example and a mistake that the idea helps you avoid.

Look for: Record scores and activations in dependency order

Reference example, not the only acceptable wording: Teacher explanation: z = [-3, 3], h = [0, 3], y = -5.

Without opening the old lesson, explain one key idea from Activation functions. Give an example and a mistake that the idea helps you avoid.

Look for: Trace identity ReLU and sigmoid on a numerical score

Reference example, not the only acceptable wording: Teacher explanation: Identity gives -2, 0, 2; ReLU gives 0, 0, 2. ReLU retains the positive score itself; it is not a step classifier.

Without opening the old lesson, explain one key idea from Representing **input**s as vectors. Give an example and a mistake that the idea helps you avoid.

Look for: Encode a record using a fixed feature order and units

Reference example, not the only acceptable wording: Teacher explanation: A [4, 2], B [2, 4], C [3, 3]. Each has dimension 2. The table has 3 rows and 2 columns.

Pictured end goal

By the end, I can explain **parameter**s and **architecture** using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Forward passes by hand

Activation functions

Representing inputs as vectors

Teach counting through one drawn unit before using a formula. Explicitly state the dense connectivity and bias assumptions.

Architecture is chosen by a designer; this does not imply it can never be selected using a separate search procedure.

Outcomes

Distinguish architectural choices from **parameter** values

Count weights and biases in a small dense network

Explain why **parameter** count alone cannot establish quality

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Make each parameter slot and counting assumption visible

> Distinguish model values,architecture,computed activations and training settings.

Main explanation,complete slots and changed-width count

Consider a dense network with 2 **input**s, 3 hidden units using ReLU, and 1 **output** unit using identity. We call its widths 2 to 3 to 1. These widths, connectivity and activation choices describe its **architecture**.

The weights and biases supply the numerical values used by that **architecture**.

A fully connected hidden unit receiving 2 **input**s needs 2 weights plus 1 bias. Three such units need 6 weights and 3 biases. The **output** receives 3 activations and needs 3 weights plus 1 bias. Total: 13 **parameter**s.

Input values and computed activations are not additional **parameter**s.

This counting rule assumes one separate bias per **output** unit and no shared or tied weights. Other **architecture**s can use different rules.

For a **dense layer** with n **input** components and m **output** units, each **output** has n weights and one bias. The count is n times m plus m, or (n + 1) times m. Count each layer separately before adding.

The helper assumes positive integer widths and the bias convention stated above. It is a counting aid, not a model trainer. A ReLU has no fitted numerical **parameter**s in this example.

Parameter slots; one value per named slot	Count
Hidden unit 1:w11, w12, b1	3
Hidden unit 2:w21, w22, b2	3
Hidden unit 3:w31, w32, b3	3
Output:v1, v2, v3, c	4
Total; inputs and hidden activations excluded	13

Same2-input/1-output dense convention · Hidden weights · Hidden biases · Output weights · Output bias · Total

> **Same2-input/1-output dense convention:** Hidden width 3
>
> **Hidden weights:** 6
>
> **Hidden biases:** 3

> **Output weights:** 3
>
> **Output bias:** 1
>
> **Total:** 13

> **Same2-input/1-output dense convention:** Hidden width 4
>
> **Hidden weights:** 8
>
> **Hidden biases:** 4
>
> **Output weights:** 4
>
> **Output bias:** 1
>
> **Total:** 17

> **Same2-input/1-output dense convention:** What one added hidden unit changes
>
> **Hidden weights:** +2
>
> **Hidden biases:** +1
>
> **Output weights:** +1
>
> **Output bias:** +0
>
> **Total:** +4

Guided: hidden layer 12, **output** layer 5, total 17, four more than 13. Exit: ask for **architecture**, activations, ordered **input** schema and exact weights/biases; equal **parameter** counts do not establish matching computations.

Complete pictured architecture and scope

Count trainable weights and biases separately from activation functions.

Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.

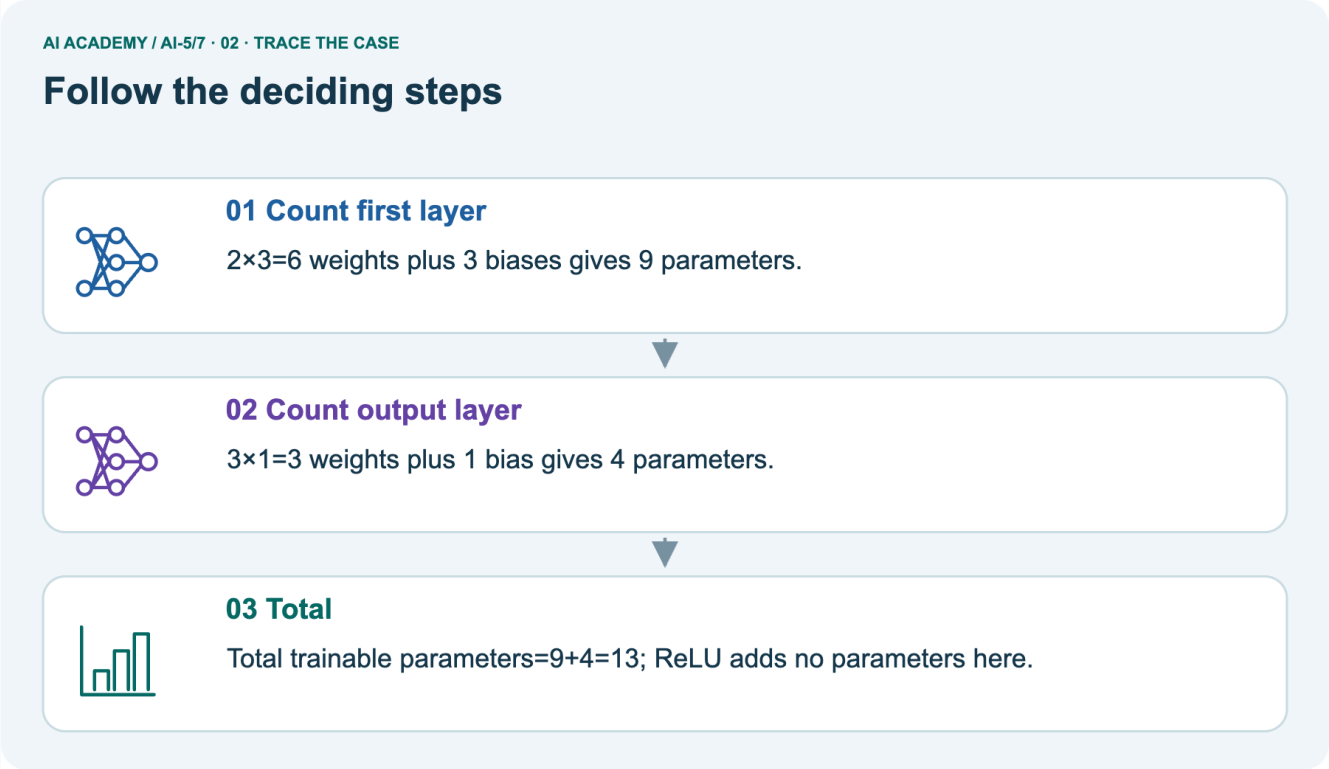
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture and main count use the same fully connected 2→3→1 architecture:6 hidden weights+3 hidden biases+3 output weights+1 output bias=13. ReLU contributes no fitted parameter here. Increasing only hidden width 3→4 adds2 incoming weights, 1 hidden bias

and 1 outgoing weight:4 more, total 17. It changes both adjacent layers; the output bias remains one.

Parameter values, computed activations and learning rate are not additional architectural connection slots.

Trace incoming,bias and outgoing increments



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Count first layer	2×3=6 weights plus 3 biases gives 9 parameters.
2. Count output layer	3×1=3 weights plus 1 bias gives 4 parameters.
3. Total	Total trainable parameters=9+4=13; ReLU adds no parameters here.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Count one weight for each **input**-to-unit connection and one bias for each receiving unit. Two **input**s connecting to three hidden units make six weights.

The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

Architecture determines the connection pattern; learned values occupy those connections.

Parameter count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

Change: Increase hidden width from 3 to4.

Prediction: First layer 12 **parameter**s; **output** layer 5; total 17.

Reason: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Parameter	A numerical model value that a fitting procedure may adjust
Architecture	The arrangement of layers, connections and activation choices
Dense layer	A layer in which each output unit has a weight for every input component
Hyperparameter	A setting chosen for a model or training procedure rather than fitted as an ordinary weight

Decide what a change alters

PROPOSED CHANGE	TYPE OF CHANGE	CONSEQUENCE
Change one weight 2 to 3	Parameter value	Same architecture
Use 4 hidden units	Architecture width	Different parameter count
Use learning rate 0.1	Training setting	No extra model weight

Each row is a separate proposed change. A parameter value, an architectural choice and a training setting answer different questions about the system.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Decide what a change alters

PROPOSED CHANGE	TYPE OF CHANGE	CONSEQUENCE
Change one weight 2 to 3	Parameter value	Same architecture
Use 4 hidden units	Architecture width	Different parameter count
Use learning rate 0.1	Training setting	No extra model weight

Each row is a separate proposed change. A parameter value, an architectural choice and a training setting answer different questions about the system.

For a dense 2 to 4 to 1 network with one bias per unit, count each layer and the total. How many more parameters does it have than 2 to 3 to 1?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Parameters and architecture

The facts we will use

AI ACADEMY / AI-5/7

Trainable parameter count

	weights	biases	total
input to hidden	6	3	9
hidden to output	3	1	4

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/7

Parameters and architecture

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Count first layer

2 x 3=6 weights plus 3 biases gives 9 parameters.



2. Count output layer

3 x 1=3 weights plus 1 bias gives 4 parameters.



3. Total

Total trainable parameters=9+4=13; ReLU adds no parameters here.

CHANGE ONE THING

Increase hidden width from 3 to4.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, **parameter**, **architecture** choice or **training** setting.

Teacher check: Teacher explanation: Activation 0.7 is computed; weight -2 is a **parameter**; width 3 is an **architecture** choice; learning rate 0.1 is a **training** setting.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Parameter**s and **architecture**. Proposed explanation: “Count one bias per **input**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Here each **output** unit has its own bias. A layer with m **output**s has m biases.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Four. The total changes from 13 to17, while the single **output** bias is not duplicated. Changed-case check: First layer 12 **parameter**s; **output** layer 5; total 17.

Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Changed-case reasoning: First layer 12 **parameter**s; **output** layer 5; total 17. Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use a connection grid and a separate bias box per **output**. Extension: for 2 to h to 1, derive $3h + h + 1 = 4h + 1$; test $h = 1$ and $h = 5$. Explain why this rule is specific to the stated **architecture**.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve fixed parameters versus changing activations from a forward pass.
7 to 18 minutes	Draw three hidden units, each with two input weights and a bias. Count without a formula first.
18 to 32 minutes	Complete W1 and the 13-parameter worked table. Derive (n + 1) times m.
32 to 45 minutes	Trace the optional counting code, classify changes and complete W2.
Session 2 start to 8 minutes	Recall the assumptions behind the count, then discuss omitted biases in W3.
8 to 20 minutes	Complete the guided 2 to 4 to 1 count; compare size without assuming quality.
20 to 35 minutes	Collect W4 independently with separate layer totals and the wider variant.
35 to 45 minutes	Use W5 and the exit response to define the evidence needed for a model comparison.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main13, wider 17, no-bias 9. Quantity sorting as supplied; architecture width is also a design hyperparameter but not a numerical model weight. Independent3→2→2:8+6=14; hiddenwidth 4 gives16+10=26. Activations are computed, not independently fitted slots. Fair quality check fixes task/schema/data/splits/evaluation and records training settings/budget/seeds; do not infer benefit from 26 versus14 alone.

Diagnose the current misconceptions

Count one bias per **input**: Here each **output** unit has its own bias. A layer with m **output**s has m biases.

Count the activation values as **parameter**s: Activations change with the **input**; the network computes them.

More weights prove better learning: Capacity and performance differ. Compare measured outcomes with controlled conditions.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

If hidden width becomes 4, recount and explain why activation values are excluded.

Teacher explanation: Original: 3 times 2 + 2 = 8, then 2 times 2 + 2 = 6, total 14. Wider: 3 times 4 + 4 = 16, then 4 times 2 + 2 = 10, total 26.

Activations are computed from **input**s and **parameter**s, not separately fitted values.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.

Dimension	Evidence required
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Count a single layer

A **dense layer** has 4 **input**s and 2 **output**s with one bias per **output**. Count weights, biases and total.

Teacher explanation: There are 8 weights and 2 biases, total 10.

W2 Sort the quantities

Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, **parameter**, **architecture** choice or **training** setting.

Teacher explanation: Activation 0.7 is computed; weight -2 is a **parameter**; width 3 is an **architecture** choice; learning rate 0.1 is a **training** setting.

W3 Change the bias convention

Recount the 2 to 3 to 1 model if every bias is omitted. State what the original count assumed.

Teacher explanation: There are 6 + 3 = 9 weights and no biases. The original 13 assumed three hidden biases plus one **output** bias.

W5 Limit a size claim

A team says its 26-**parameter** model must outperform a 14-**parameter** model. Give a fair check and one reason the claim is unsupported.

Teacher explanation: Compare on the same valid evaluation task and cases after a documented **fitting process**. Count alone does not establish suitable features, successful optimisation or generalisation.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: hidden layer 12, **output** layer 5, total 17, four more than 13. Exit: ask for **architecture**, activations, ordered **input** schema and exact weights/biases; equal **parameter** counts do not establish matching computations.

W1 Count a single layer

W2 Sort the quantities

W3 Change the bias convention

W4 Count a new architecture independently

A dense 3 to 2 to 2 network uses ReLU in the hidden layer and identity in the **output** layer, with one bias per unit. Give each layer count and the total.

If hidden width becomes 4, recount and explain why activation values are excluded.

Teacher explanation: Original: 3 times 2 + 2 = 8, then 2 times 2 + 2 = 6, total 14. Wider: 3 times 4 + 4 = 16, then 4 times 2 + 2 = 10, total 26.

Activations are computed from **input**s and **parameter**s, not separately fitted values.

W5 Limit a size claim

Independent assessment and fresh retry

W4 must show 8 + 6 = 14 and 16 + 10 = 26. Fresh retry: dense 2 to 2 to 3 with biases has 6 + 9 = 15 **parameter**s. Without biases it has 4 + 6 = 10. Require the bias convention explicitly.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled **parameter**s and **architecture evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish architectural choices from **parameter** values.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use a connection grid and a separate bias box per **output**. Extension: for 2 to h to 1, derive 3h + h + 1 = 4h + 1; test h = 1 and h = 5. Explain why this rule is specific to the stated **architecture**.

Week 8 introduces losses for comparing **prediction**s with targets.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/8, ai-5/10, ai-5/13. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Distinguish architectural choices from **parameter** values

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Classify hidden activation 0.7, weight -2, hidden width 3 and learning rate 0.1 as computed value, **parameter**, **architecture** choice or **training** setting.

Reasoned practice answer: Teacher explanation: Activation 0.7 is computed; weight -2 is a **parameter**; width 3 is an **architecture** choice; learning rate 0.1 is a **training** setting.

Check every learner before discussion: Use the worked case for **Parameter**s and **architecture**. Proposed explanation: “Count one bias per **input**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Here each **output** unit has its own bias. A layer with m **output**s has m biases.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Increase hidden width from 3 to4. Require a **prediction** and a reason before comparing. Expected result: First layer 12 **parameter**s; **output** layer 5; total 17.

Reason: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

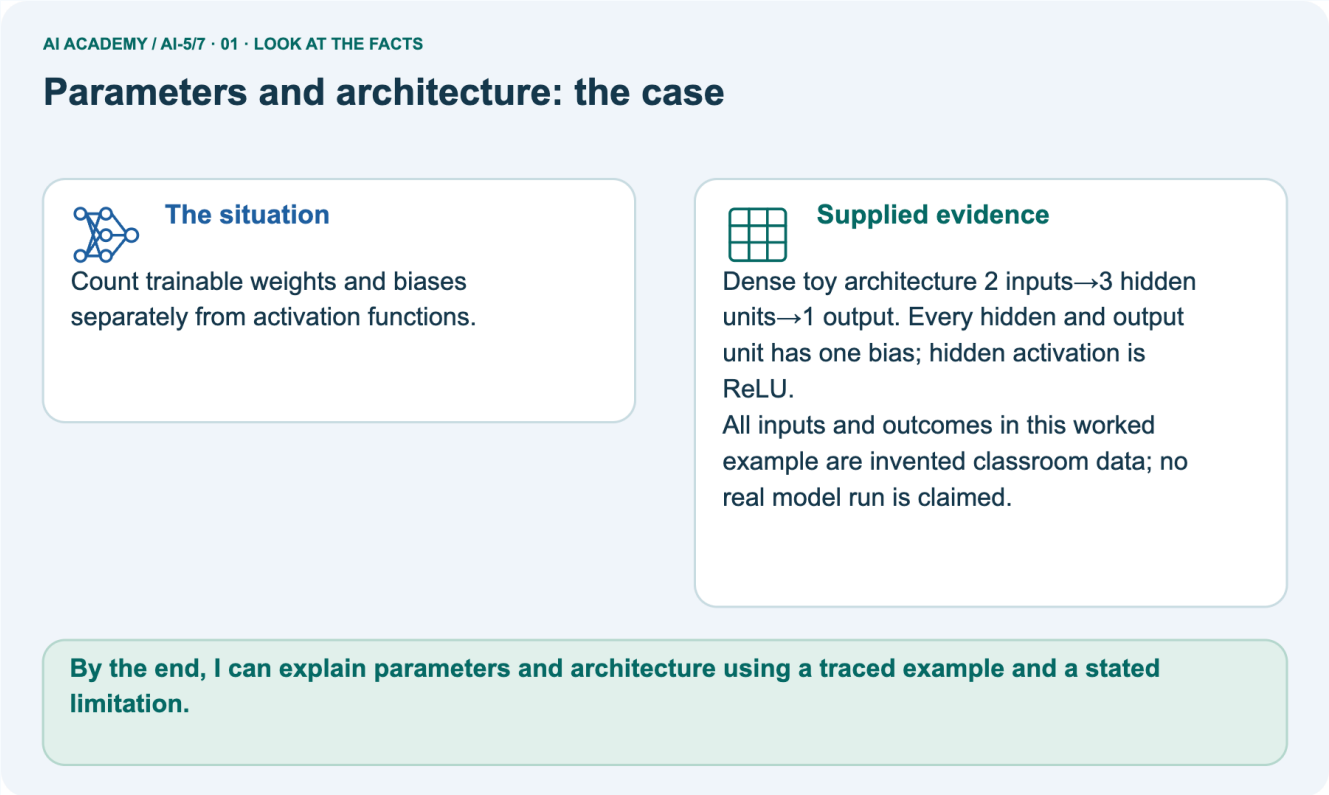
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Parameters and architecture: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

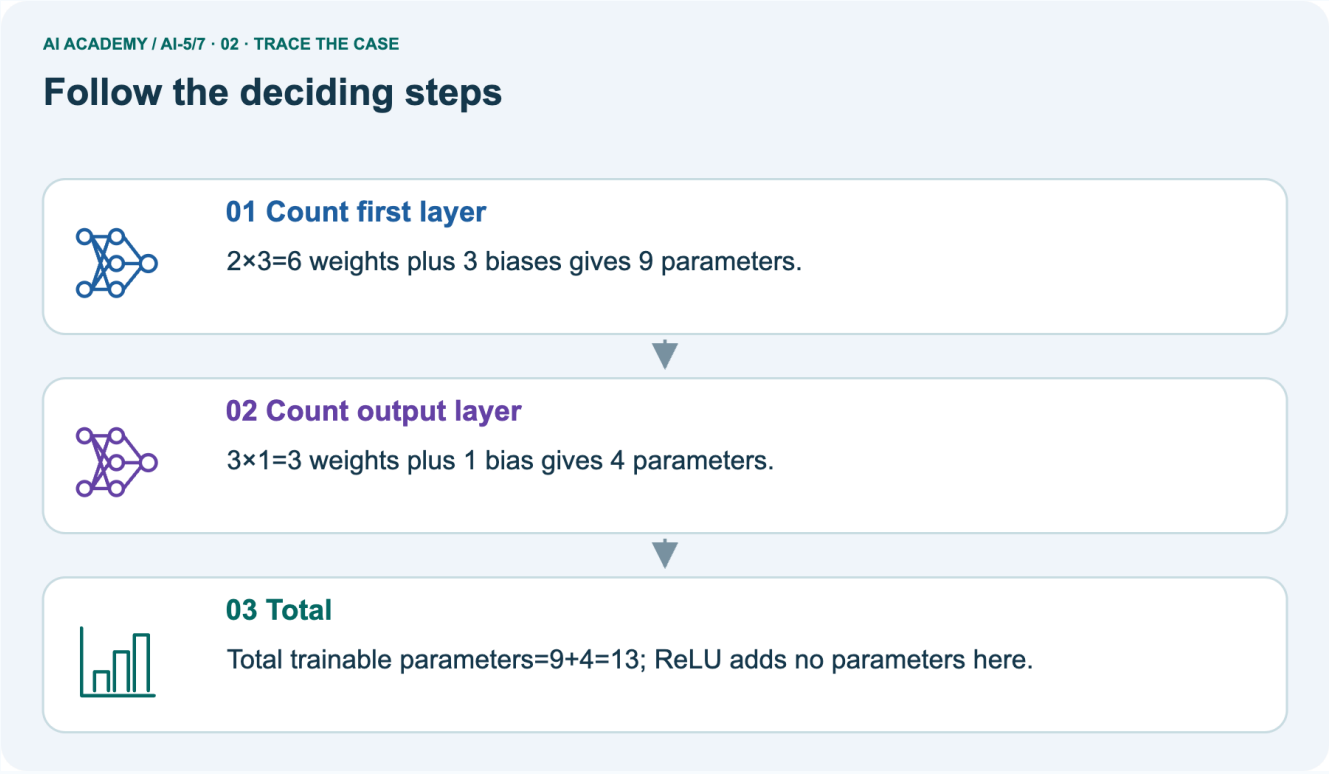
Count trainable weights and biases separately from activation functions. Supplied facts: Dense toy **architecture** 2 **input**s→3 hidden units→1 **output**. Every hidden and **output** unit has one bias; hidden activation is ReLU.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



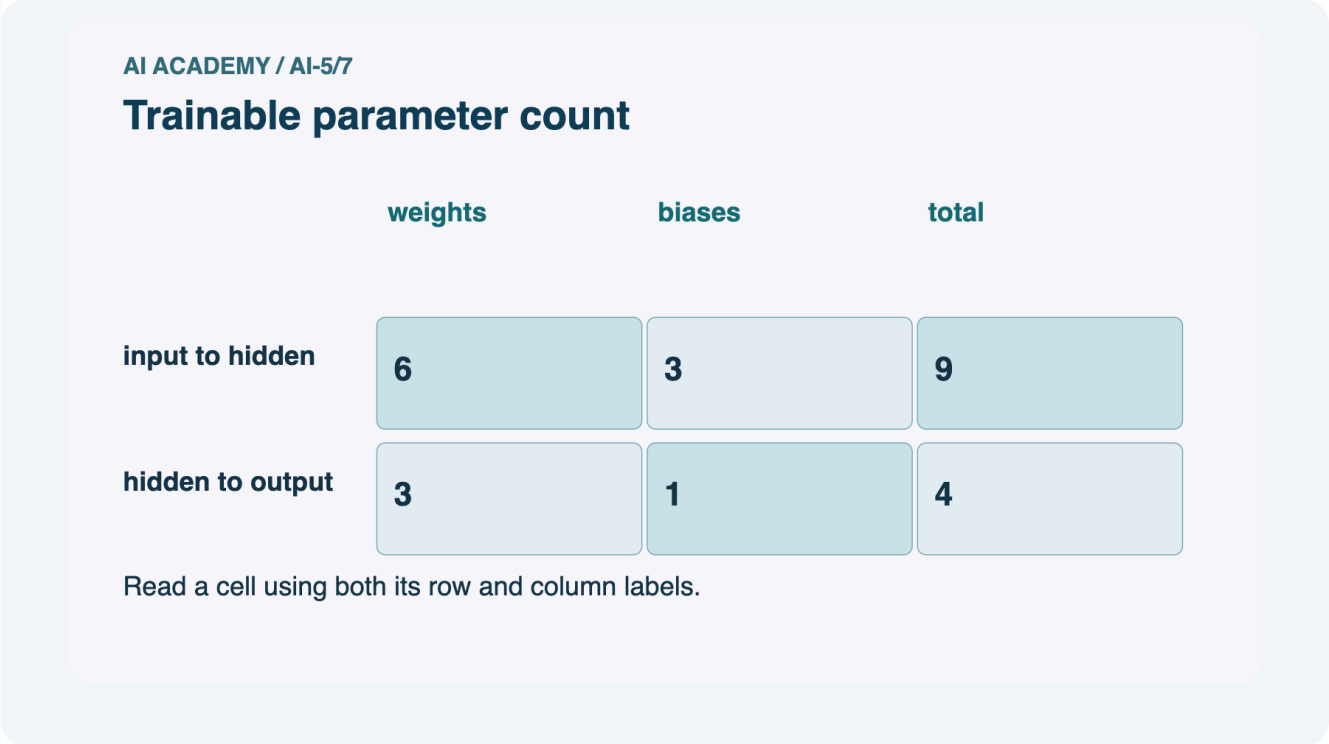
Follow the labelled picture with the matching explanation. Open the vector diagram

Count first layer: 2×3=6 weights plus 3 biases gives 9 **parameter**s. → Count **output** layer: 3×1=3 weights plus 1 bias gives 4 **parameter**s. → Total: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Trainable parameter count



Follow the labelled picture with the matching explanation. Open the vector diagram

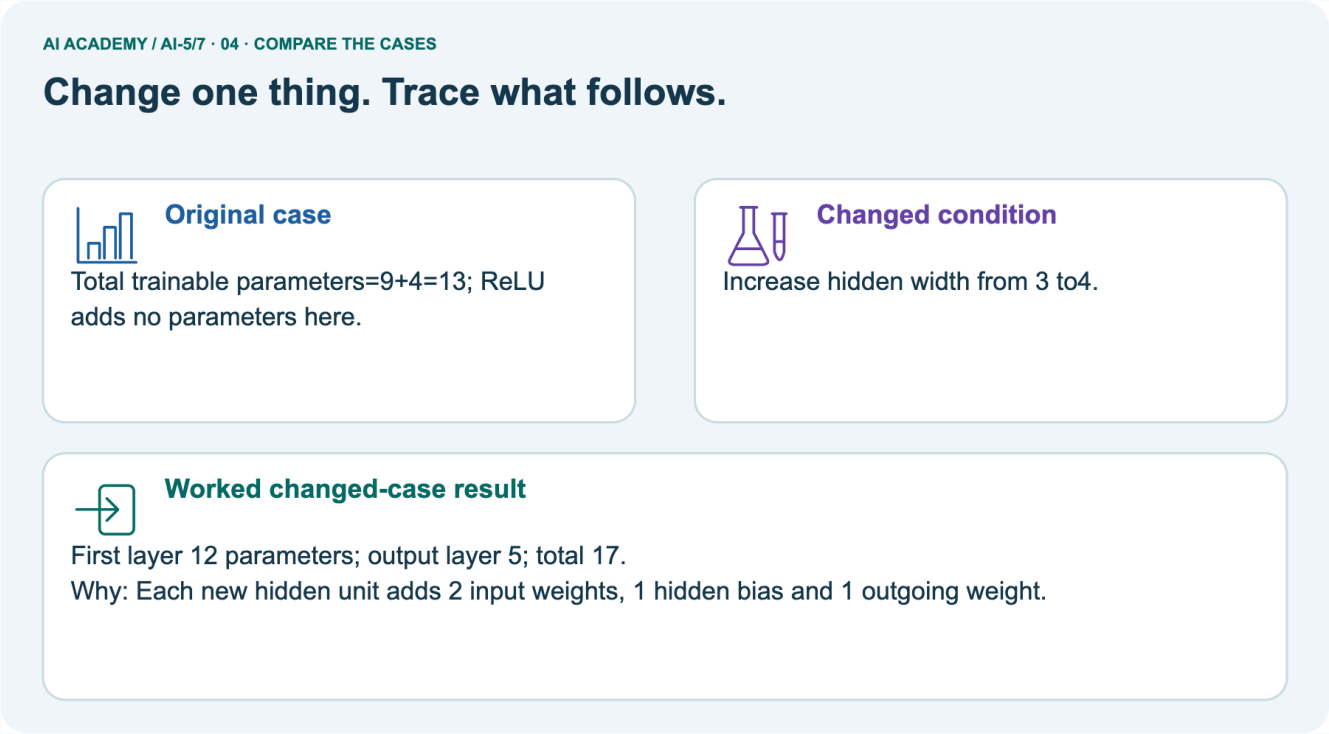
Trainable **parameter** count. **input** to hidden: weights = 6, biases = 3, total = 9. hidden to **output**: weights = 3, biases = 1, total = 4.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Total trainable **parameter**s=9+4=13; ReLU adds no **parameter**s here. Changed condition: Increase hidden width from 3 to4. Expected result: First layer 12 **parameter**s; **output** layer 5; total 17.

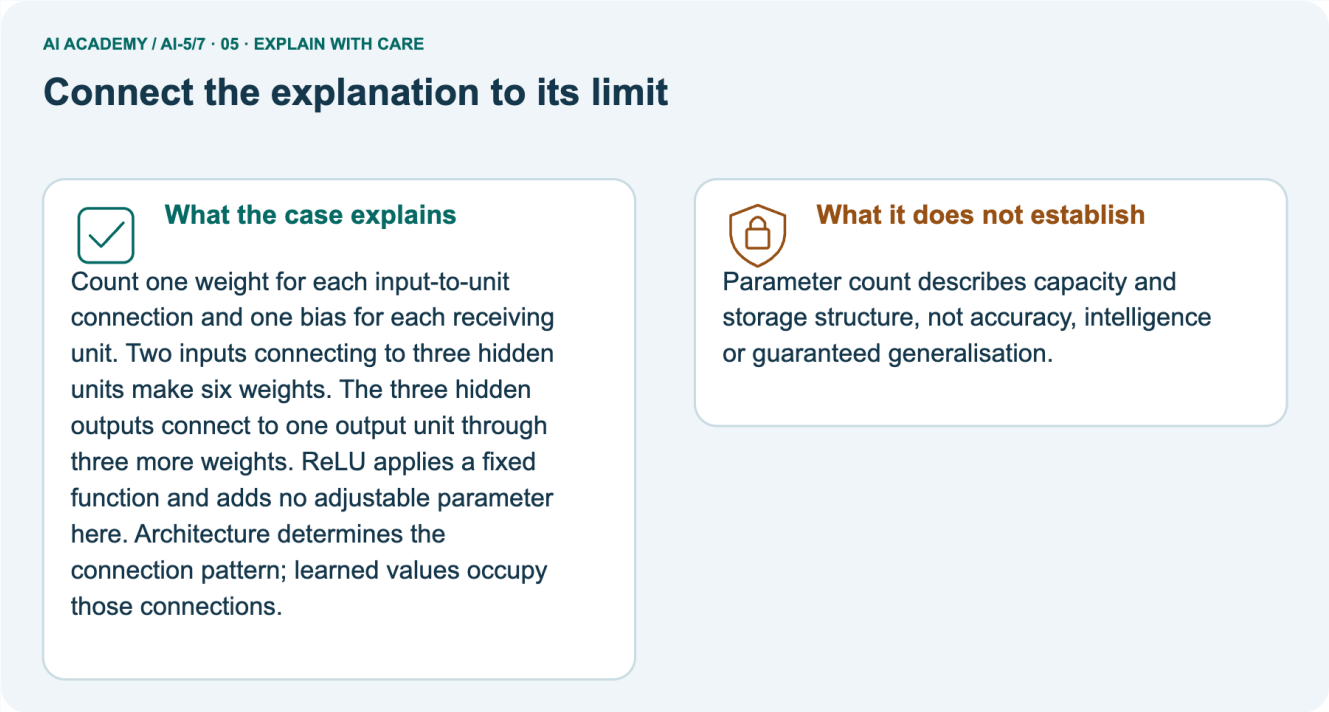
Reason: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Count one weight for each **input**-to-unit connection and one bias for each receiving unit. Two **input**s connecting to three hidden units make six weights.

The three hidden **output**s connect to one **output** unit through three more weights. ReLU applies a fixed function and adds no adjustable **parameter** here.

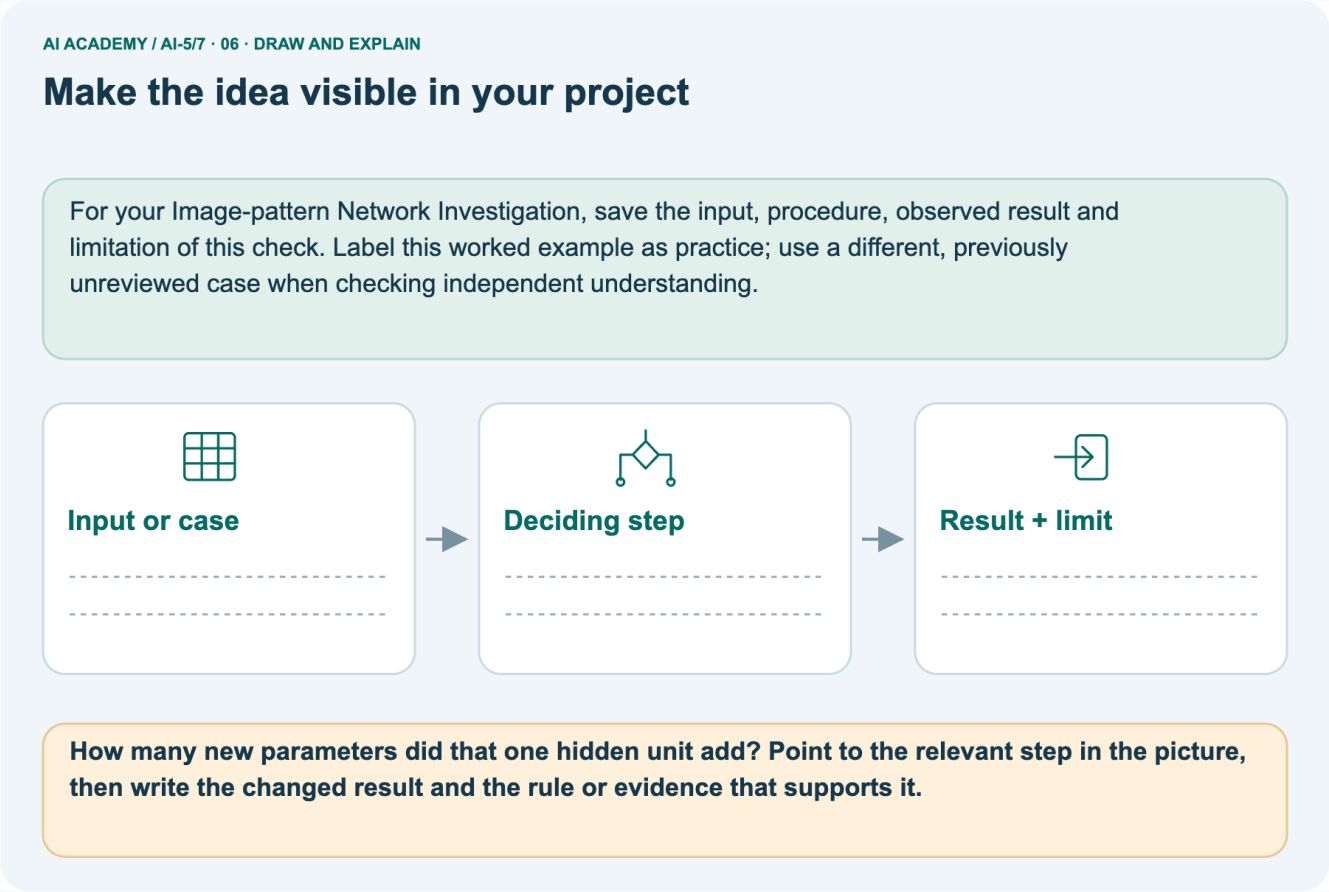
Architecture determines the connection pattern; learned values occupy those connections. Limit: **Parameter** count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Parameter** count describes capacity and storage structure, not accuracy, intelligence or guaranteed generalisation.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many new **parameter**s did that one hidden unit add? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: Four. The total changes from 13 to17, while the single **output** bias is not duplicated. Changed-case check: First layer 12 **parameter**s; **output** layer 5; total 17.

Each new hidden unit adds 2 **input** weights, 1 hidden bias and 1 outgoing weight.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Parameters and architecture: the case

02. Follow the deciding steps

03. Trainable parameter count

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.